

Technology Stack

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend/ Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Used to build a responsive, user-friendly, and interactive web interface for farmers.

2	Backend/Server-Side	Python (Flask)	Flask provides a lightweight web framework to handle user requests and integrate the machine learning model efficiently.
3	Database / Data Storage	SQLite	A lightweight relational database used to store user information, login details, and prediction history.
4	Cloud / Hosting / Deployment	Localhost (Flask Development Server)	Used for development, testing, and demonstration of the project. The application can later be deployed to cloud platforms if required.
5	Version Control & CI/CD	GitHub	Used for source code management, version control, and collaboration among team members.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib	Scikit-learn is used to train the crop prediction model, Pandas and NumPy are used for data processing, and Joblib is used to save and load the trained model.